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\documentclass{article}
\usepackage{amsmath, amssymb, amsthm, bm, amsfonts}
\usepackage{geometry}
\geometry{a4paper, margin=1in}
\usepackage{hyperref}
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\begin{document}
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\title{Development of a Floer Differential Operator within the Multiplicity Framework}
\author{}
\date{}
\maketitle
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\section*{Overview}
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To adapt Floer differential operators to the principles of Multiplicity Theory, we extend the classical symplectic framework to include:

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\begin{itemize}
  \item \textbf{Multiplicity Dynamics}: Integration of eigenvalue-based interactions and recursive feedback.
  \item \textbf{Tensor Networks}: High-dimensional state representations to capture multi-scale interactions.
  \item \textbf{Stochastic Feedback}: Modeling environmental noise and dynamic fluctuations.
  \item \textbf{Prime-Based Encoding}: Modular and unique representations of trajectories.
\end{itemize}
```

The resulting operator, denoted $\mathcal{F}$, incorporates these elements to study advanced dynamics in high-dimensional multiplicative systems.

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\section*{Extended Floer Differential Operator}
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The extended operator $\mathcal{F}$ is defined as:

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\begin{equation}
  \mathcal{F}(u) = \frac{\partial u}{\partial t} + J \nabla H(u) + \sum_{i,j} T_{ij} \cdot \nabla \Phi(u) + \xi(t),
\end{equation}
```

where:

```
\begin{itemize}
  \item  $\nabla H(u)$ : Gradient flow of the Hamiltonian.
  \item  $T_{ij}$ : Tensor coefficients representing multi-scale interactions.
  \item  $\nabla \Phi(u)$ : Feedback-adjusted potential gradient.
  \item  $\xi(t)$ : Stochastic term modeling noise.
\end{itemize}
```

Enhancements include:

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\begin{itemize}
  \item \textbf{Nonlinearity}: Self-interaction term  $\eta u^2$ .
  \item \textbf{Dynamic Parameters}: Time-dependent coefficients,  $\alpha_k(t)$ ,  $\beta_k(t)$ .
  \item \textbf{Prime Encoding}: Mapping  $u_i \sim p_i$ , where  $p_i$  are primes.
\end{itemize}
```

\end{itemize}

\section*{Applications and Mathematical Extensions}

\subsection*{1. Topological Quantum Field Theory (TQFT)}

Floer trajectories contribute to topological invariants via tensor representations:

\begin{equation}

$$\Psi(u) = \sum_{i,j} T_{ij} \cdot u_i \otimes u_j \cdot e^{i(\theta_i - \theta_j)}.$$

\end{equation}

These invariants, such as the Euler characteristic $\chi(M)$, can be refined using:

\begin{equation}

$$\chi(M) = \int_M \left(R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} - 4R_{\mu\nu} R^{\mu\nu} + R^2 \right) d^4x.$$

\end{equation}

\subsection*{2. Hybrid Quantum-Classical Systems}

Floer dynamics integrate quantum corrections for hybrid modeling:

\begin{equation}

$$\mathcal{F}(u) = \frac{\partial u}{\partial t} + J \nabla H(u) + \sum_{m,n} \gamma_{mn} u_m u_n \cos(\theta_m - \theta_n).$$

\end{equation}

Here, γ_{mn} are interaction weights derived from quantum coherence terms.

\subsection*{3. Adaptive Learning Systems}

Recursive feedback enhances adaptability in AI models:

\begin{equation}

$$\mathcal{L}(t) = \sum_{i=1}^N \nabla H(u_i) \cdot \cos(\omega_i t + \phi_i) \cdot F(t) \cdot (1 + \epsilon_i(t)).$$

\end{equation}

The feedback function $F(t)$ dynamically adjusts weights based on prior states.

\section*{Conclusion}

The extended Floer differential operator integrates classical dynamics with multiplicative principles, enabling new insights in quantum systems, topological field theories, and adaptive computation. Future work will explore numerical validations and interdisciplinary applications.

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